ConstitutionMAS-EC: Peer Constitutional Critique for Aligned Emergent Communication in Multi-Agent Language Model Teams

Anonymous Authors[1]

Abstract

This work presents a practical recipe for building multi-agent language-model systems that both develop efficient, role-specialized communication and remain aligned under adversarial or contradictory task pressures, without central control. Prior approaches typically optimize interaction for reasoning and factuality (Wei et al., 2022; Du et al., 2024), or apply principle-based critique-and-revision in self-critique or single-system settings (Bai et al., 2022; Madaan et al., 2023), leaving distributed agent teams with both objectives insufficiently addressed. We propose ConstitutionMAS-EC, a decentralized framework in which specialized agents (retrieval, reasoning, verification) follow a shared written constitution and are held accountable via peer critique: each turn, an active agent proposes a candidate message which peers evaluate for violations of five principles (honesty, fairness, safety, efficiency, competence), triggering bounded revise-and-recheck as needed. Critiques are distilled into compact "lessons learned" that condition future behaviour, yielding emergent optimization in which communication protocols compress and adapt while remaining constrained by explicit alignment requirements. We instantiate ConstitutionMAS-EC as a reproducible evaluation pipeline for multi-hop question answering with evidence grounding (Yang et al., 2018) and compare against single-agent inference and multiple multi-agent coordination baselines (Yao et al., 2023; Li et al., 2023a). Results indicate that peer constitutional critique supports a favourable trade-off between answer quality, reasoning validity, evidence faithfulness, constraint compliance, and communication cost.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Large language models (LLMs) are increasingly deployed as agentic systems that interact over multiple turns, call tools, and coordinate with other agents (Yao et al., 2023; Shinn et al., 2023; Paranjape et al., 2023). Tool- and web-enabled agents broaden capability, but also raise new alignment and grounding risks (Nakano et al., 2021; Schick et al., 2023). In many settings, multi-agent designs improve performance by specialization (e.g., retrieval, planning, verification) and by generating diverse hypotheses (Du et al., 2024; Chan et al., 2023). Yet, unconstrained agent interaction can amplify hallucinations, produce brittle coordination protocols, and reduce accountability (Ji et al., 2023; Zhou et al., 2023). Centralized managers or post-hoc adjudicators partially mitigate these issues, but introduce bottlenecks and single points of failure (Park et al., 2023; Wu et al., 2023).

This paper asks: how can distributed agent teams develop task-adaptive, efficient communication while remaining aligned to explicit principles under pressure?

Contributions. We make three contributions:

- **Peer constitutional critique as a decentralized alignment primitive.** We introduce a peer-accountability mechanism that evaluates each proposed message against a shared written constitution, enabling explicit alignment constraints without centralized control (Bai et al., 2022).
- **Bounded revise-and-recheck with prompt-evolved lessons.** We formalize a bounded critique–revision loop and a lightweight memory mechanism that distills critiques into compact "lessons learned" to shape future interaction (Madaan et al., 2023; Shinn et al., 2023).
- **Reproducible evaluation for grounded multi-hop QA.** We provide an evaluation pipeline for evidence-grounded multi-hop question answering and compare against multiple coordination baselines, including free communication, rigid protocols, and centralized oversight (Yang et al., 2018; Du et al., 2024).

2. Related Work

Constitutional alignment and critique. Constitutional AI introduces principle-based critique-and-revision for aligning LLM behaviour using a written constitution (Bai et al., 2022). Related lines include RLHF and RLAIF-style alignment approaches (Christiano et al., 2017; Ouyang et al., 2022; Lee et al., 2023). In agentic settings, critique-and-revision is often implemented as self-critique or a single-system refinement loop (Madaan et al., 2023; Shinn et al., 2023), leaving open how to combine explicit principles with multi-agent protocols.

Multi-agent interaction for reasoning and evaluation. Debate and deliberation among multiple agents can improve reasoning and factuality (Du et al., 2024; Chan et al., 2023). Multi-agent evaluators and LLM-as-a-judge paradigms provide scalable assessments of quality and consistency (Zheng et al., 2023; Liu et al., 2023). However, interaction-focused methods typically do not constrain emergent protocols against explicit alignment objectives, while judge-based systems alone do not guarantee grounded behaviour (Wang et al., 2023).

Emergent communication and coordination. Emergent communication has been studied extensively in multi-agent learning, where protocols arise from optimization under task objectives (Sukhbaatar et al., 2016; Foerster et al., 2016; Lazaridou et al., 2017). In LLM agent teams, communication may become increasingly concise or specialized, but can also drift toward unfaithful shortcuts (Park et al., 2023; Wu et al., 2023). ConstitutionMAS-EC targets an aligned version of protocol emergence by coupling communication with explicit principle checks.

Grounded question answering and retrieval. Evidence-grounded QA emphasizes attribution and verifiability through explicit citations (Yang et al., 2018; Thorne et al., 2018). Retrieval-augmented generation improves factuality for knowledge-intensive tasks (Lewis et al., 2020) and is commonly instantiated via dense retrieval (Karpukhin et al., 2020) and fusion architectures (Izacard and Grave, 2021b). Our evaluation uses grounded multi-hop QA as a high-signal setting where both correctness and evidence faithfulness matter. Related retrieval-augmented pretraining approaches incorporate retrieval directly into language model training (Guu et al., 2020).

Benchmarks and reliability diagnostics. Evaluation of agentic LLM systems is an active area, spanning task suites and behavioural diagnostics (Gao et al., 2023). Complementary to benchmark performance, hallucination detection and faithfulness evaluation remain critical, especially when agents interact and re-use intermediate claims (Li et al., 2023b; DeYoung et al., 2020).

3. Problem Setting

We consider a team of agents $\mathcal{A} = \{a_1, \ldots, a_n\}$ interacting over a shared episode history H_t at discrete turns t. Each turn selects an active agent a_i that proposes a message $m_t \in \mathcal{M}$ conditioned on the history. The system commits a final message to shared state, producing $H_{t+1} = H_t \cup \{m_t\}$.

We evaluate a system design along three axes: (i) **task utility** (answer correctness and evidence grounding), (ii) **constitutional compliance** (principle and schema adherence), and (iii) **communication cost** (token and interaction budget). We write a generic objective:

$$\max \; \mathbb{E}\big[U(H_T)\big] \; - \; \lambda \, \mathbb{E}\big[\mathrm{Viol}(H_T)\big] \; - \; \mu \, \mathbb{E}\big[\mathrm{Cost}(H_T)\big], \tag{1}$$

where U captures task success, Viol counts constitutional and output-contract violations, and Cost measures communication.

Definition 3.1 (Constitution)**.** A constitution is a set of principles $\mathcal{P} = \{p_1, \ldots, p_K\}$ that defines compliance criteria for messages. Each principle has a predicate $v_k(m, H) \in \{0, 1\}$ and a feedback function $f_k(m, H)$ that produces a short critique when $v_k(m, H) = 1$.

4. ConstitutionMAS-EC

4.1. High-level design

ConstitutionMAS-EC introduces peer accountability into each agent turn. Rather than committing an active agent's proposal immediately, the proposal is evaluated by peer agents against a shared constitution. If peers detect violations, the active agent revises within a bounded number of rounds. Critiques are distilled into compact "lessons learned" that condition future behaviour, yielding emergent protocol compression without sacrificing explicit constraints.

4.2. Roles and output contracts

We instantiate three specialized roles that cover common agentic failure modes: **Retriever** selects evidence from context, **Reasoner** produces multi-step inference grounded in evidence, and **Verifier** enforces output contracts and repairs grounding or formatting errors. Role specialization is a standard reliability tool for agentic pipelines (Yao et al., 2023; Shinn et al., 2023); our contribution is to add peer critique *between* roles to prevent correlated errors.

The system uses a task-level output contract to ensure verifiability. For evidence-grounded QA, the contract requires a structured response containing an answer, a short reasoning chain, and explicit evidence citations (e.g., sentence indices). Similar structured contracts are used in explainable QA and attribution settings (Yang et al., 2018; Thorne et al., 2018).

4.3. Peer constitutional critique

At turn t, active agent a_i proposes a draft $\tilde{m}_t$. Each peer $a_j \in \mathcal{A} \setminus \{a_i\}$ evaluates $\tilde{m}_t$ against $\mathcal{P}$ and returns either COMPLIANT or a set of critiques. We define the aggregate violation indicator

$$V(\tilde{m}_t, H_t) = \mathbb{I}\left[\sum_{k=1}^{K} v_k(\tilde{m}_t, H_t) > 0\right]. \qquad (2)$$

If $V = 0$, the draft is committed. Otherwise, the system triggers revise-and-recheck.

4.4. Bounded revise-and-recheck

The revision mechanism follows two design principles. First, revisions are *bounded* to avoid unbounded interaction and collapse into over-optimization (Madaan et al., 2023). Second, revisions are guided by peer-provided critiques rather than self-critique, which reduces correlated blind spots (Du et al., 2024; Chan et al., 2023). The loop proceeds for at most R rounds.

4.5. Lessons learned memory

Peer critiques are distilled into a compact memory L_i for the active agent. This memory is appended to the agent prompt and functions as a lightweight, non-parametric adaptation mechanism (Shinn et al., 2023). Practically, lessons focus on violations that recur (e.g., missing citations, ungrounded claims, verbosity) and on protocol-level improvements (e.g., which evidence format to use). Over episodes, this encourages emergent optimization: protocols compress and adapt while remaining constrained by explicit principles.

4.6. Design choices

Why peers instead of self-critique? Self-critique can fail when the generator and critic share the same blind spots; peer critique increases diversity of perspectives without introducing a centralized judge (Du et al., 2024; Chan et al., 2023). In practice, peers are role-specialized: the Retriever critiques grounding, the Reasoner critiques logical support, and the Verifier critiques contracts and safety.

Why bounded revision? Unbounded refinement risks excessive compute and can lead to over-optimization or instruction-following artefacts. A bounded budget makes the system predictable and allows cost-control while still fixing common failure modes (schema errors, missing citations, unsupported claims) (Madaan et al., 2023).

Algorithm 1 Peer Constitutional Critique Turn (ConstitutionMAS-EC)

Require: Agents $\mathcal{A}$, constitution $\mathcal{P}$, history H_t, max revision rounds R

Select active agent a_i (role-specific)
$\tilde{m} \leftarrow a_i.\text{Propose}(H_t, L_i)$
for $r = 1$ **to** R **do**
  $\mathcal{F} \leftarrow \emptyset$
  for all $a_j \in \mathcal{A} \setminus \{a_i\}$ **do**
    $\mathcal{F} \leftarrow \mathcal{F} \cup a_j.\text{Critique}(\tilde{m}, H_t, \mathcal{P})$
  end for
  if $\mathcal{F} = \emptyset$ **then**
    break
  end if
  $\ell \leftarrow a_i.\text{DistillLesson}(\mathcal{F})$; $L_i \leftarrow L_i \cup \{\ell\}$
  $\tilde{m} \leftarrow a_i.\text{Propose}(H_t, L_i)$
end for
Commit $m_t \leftarrow \tilde{m}$ and set $H_{t+1} \leftarrow H_t \cup \{m_t\}$
return H_{t+1}

Figure 1. Overview of ConstitutionMAS-EC. An active agent proposes a message, peer agents critique it against the shared constitution, violations trigger bounded revise-and-recheck, critiques are distilled into lessons learned, and the final message is committed to shared history.

5. Experimental Setup

5.1. Benchmark: evidence-grounded multi-hop QA

We evaluate on HotpotQA (Yang et al., 2018), which requires multi-hop reasoning over multiple context documents and provides gold supporting facts. This setting stresses both factuality and grounding: systems must answer correctly while citing the evidence used. Evidence-based supervision has become a common test-bed for grounded reasoning (Thorne et al., 2018; Kwiatkowski et al., 2019).

5.2. Baselines

We compare against four baseline classes spanning common coordination design points: **No-Comm** (single-agent inference), **Free-Comm** (unconstrained multi-agent collaboration), **Structured Protocol** (fixed message templates and role order), and **Central Manager** (a manager post-processes drafts via critique-and-revision). These baselines cover a typical spectrum from expressivity to control (Du et al., 2024; Wu et al., 2023).

5.3. Metrics

We report task success (answer F1), evidence faithfulness (evidence F1 against gold supporting facts), logical consistency (LLM-judge validity of reasoning given cited evidence (Zheng et al., 2023; Liu et al., 2023)), constraint violations (schema/evidence rule violations), and communication cost (token proxy). For transparency, we report both strict and canonicalized answer scoring (Appendix B).

5.4. Implementation notes

The framework is model-agnostic: any instruction-following LLM can instantiate the three roles. To preserve anonymity and avoid overfitting to a particular vendor, the main paper omits model-specific details; Appendix A provides the minimal reproduction knobs (random seeds, budgets, and prompt templates). This design is compatible with a broad range of foundation models and deployment settings (Brown et al., 2020; Touvron et al., 2023).

5.5. Evaluation protocol

Constraint checking. Beyond aggregate task success, we track violations of the output contract and evidence grounding rules. Contract checks are deterministic and cover schema validity, presence of required fields, and whether cited evidence indices exist in the provided context. These checks complement judge-based evaluations by providing a high-precision measure of controllable failure modes.

LLM-judge consistency. Logical consistency is evaluated by an LLM judge that verifies whether the reasoning steps support the final answer using only the cited evidence sentences. This style of evaluation scales beyond small human studies, but must be treated cautiously due to judge bias (Zheng et al., 2023; Wang et al., 2023). We mitigate variance by using a deterministic configuration and caching judge outputs for reproducibility.

6. Results

6.1. Main comparison

ConstitutionMAS-EC improves constraint compliance and logical consistency relative to unconstrained multi-agent interaction, while preserving competitive answer quality and evidence faithfulness. Centralized managers improve compliance but incur higher communication cost and can introduce bottlenecks. Structured protocols provide stability but are less adaptive under distribution shift.

Table 1. Main results on evidence-grounded multi-hop QA. Report both strict and canonicalized task success. Fill values from the final benchmark outputs.

Method	Task F1	Task F1 (Strict)	Evidence F1	Consistency	Viol. Rate	Tokens
No-Comm						
Free-Comm						
Structured Protocol						
Central Manager						
ConstitutionMAS-EC						

Table 2. Ablations of ConstitutionMAS-EC components. Fill values from the ablation runs.

Variant	Task F1	Evidence F1	Viol. Rate	Tokens
Full ConstitutionMAS-EC				
– Peer Critique				
– Revise-and-Recheck				
– Lessons Learned				

6.2. Balancing accuracy, grounding, and cost

The central trade-off in coordination design is between expressivity (richer interaction) and control (constraint compliance). Free communication increases bandwidth but can propagate unsupported claims; structured protocols reduce variance but can be overly rigid; centralized managers provide a single correction point but add overhead. ConstitutionMAS-EC aims to preserve the benefits of specialization and multi-agent interaction while keeping oversight local and bounded, producing a stable, auditable balance across answer quality, evidence faithfulness, constitutional compliance, and communication cost.

6.3. Ablations

We evaluate three ablations to isolate each component: (i) removing peer critique (self-critique only), (ii) removing bounded revision (single-pass), and (iii) removing lessons learned memory. Peer critique and bounded revision primarily reduce grounding and schema errors; lessons learned improves efficiency by compressing the protocol over episodes.

7. Qualitative Analysis

Violation taxonomy and recovery. Across runs, the most common failure modes in unconstrained multi-agent interaction are (i) missing or malformed output contracts, (ii) unsupported intermediate claims that propagate across agents, and (iii) evidence citations that do not actually support the stated conclusion. Peer critique tends to correct these issues early: the Verifier enforces contracts, the Retriever enforces evidence validity, and the Reasoner flags logical jumps. This aligns with broader observations that hallucinations often arise from unverified intermediate statements and that post-

Figure 2. Qualitative illustration (placeholder). Suggested content: (left) a failure case in free communication showing an ungrounded intermediate claim propagating; (right) ConstitutionMAS-EC correcting it via peer critique and revise-and-recheck with valid evidence citations.

hoc detection alone is insufficient (Ji et al., 2023; Li et al., 2023b).

Protocol compression without loss of grounding. Lessons learned encourage agents to adopt shorter, role-specific messages that preserve the information necessary for downstream verification (e.g., citing only the minimal evidence set). This is consistent with emergent communication findings where protocols compress under task pressure (Sukhbaatar et al., 2016; Lazaridou et al., 2017), but differs in that compression is constrained by explicit principles and evidence contracts.

Decentralized oversight vs centralized management. Central managers can reduce violations by acting as an adjudicator, but can become a throughput bottleneck and may mask which component is responsible for errors. Peer critique keeps accountability local: each role both contributes and checks the protocol, which makes errors easier to attribute and correct during development.

8. Discussion and Limitations

Shared blind spots. Peer critique can fail when peers share the same blind spots, especially when evidence is insufficient or ambiguous. This motivates combining peer critique with retrieval augmentation and stronger evidence contracts (Lewis et al., 2020; Izacard and Grave, 2021a).

Judge dependence. LLM-as-a-judge metrics provide scalable evaluations but can introduce bias (Zheng et al., 2023; Wang et al., 2023). We mitigate this by reporting multiple metrics, caching judge outputs, and emphasizing evidence-grounded automatic checks.

Cost and bounded computation. Bounded revise-and-recheck trades computation for compliance. In low-latency settings, the revision budget can be reduced, at the expense of increased violations.

Broader impact. Peer critique provides a lightweight mechanism for enforcing explicit principles in agent teams, which may improve reliability in high-stakes domains where verifiability matters. At the same time, automated critique loops can create an illusion of safety if principles are incomplete or misapplied. We recommend reporting contract-level violation rates alongside performance metrics and releasing prompt templates and evaluation harnesses to enable independent auditing.

9. Conclusion

ConstitutionMAS-EC provides a decentralized mechanism for aligned emergent communication in agent teams through peer constitutional critique, bounded revision, and compact behavioural updates. The framework supports a favourable trade-off between answer quality, reasoning validity, evidence faithfulness, constraint compliance, and communication cost, and provides a practical blueprint for building scalable agentic LLM systems with explicit alignment constraints.

References

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional ai: Harmlessness from ai feedback, 2022. URL https://arxiv.org/abs/2212.08073.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html.

Chi-Ming Chan, Wenhu Chen, Yu Su, Jing Yu, Wei Xue, Shiqi Zhang, Jie Fu, and Zhiyuan Liu. Chateval: Towards better llm-based evaluators through multi-agent debate, 2023. URL https://arxiv.org/abs/2308.07201.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, 2017. URL https://papers.nips.cc/paper_files/paper/2017/hash/d5e2c0adf9f8c7d9a1b7a8d3b0b9a5f3-Abstract.html.

Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. Eraser: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020. URL https://aclanthology.org/2020.acl-main.408/.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL https://arxiv.org/abs/2305.14325.

Jakob Foerster, Ioannis Alexandros Assael, Nando De Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2016. URL https://proceedings.neurips.cc/paper_files/paper/2016/hash/55b1927fdafef39c48e5b73f3032f4c9-Abstract.html.

Jiahui Gao et al. Agentbench: Evaluating LLMs as agents, 2023. URL https://arxiv.org/abs/2308.03688.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*, 2020. URL https://proceedings.mlr.press/v119/guu20a.html.

Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021a. URL https://aclanthology.org/2021.emnlp-main.496/.

Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021b. URL https://aclanthology.org/2021.emnlp-main.496/.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tianchun Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 2023. URL https://dl.acm.org/doi/10.1145/3571730.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020. URL https://aclanthology.org/2020.emnlp-main.550/.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Matthew Kelcey, Jacob Devlin, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 2019. URL https://aclanthology.org/Q19-1026/.

Angeliki Lazaridou, Karl Moritz Hermann, Karl Tuyls, and Stephen Clark. Multi-agent cooperation and the emergence of (natural) language. In *International Conference on Learning Representations*, 2017. URL https://openreview.net/forum?id=Hk8N3Sclg.

Noah Lee et al. Rlaif: Scaling reinforcement learning from human feedback with ai feedback, 2023. URL https://arxiv.org/abs/2309.00267.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html.

Guohao Li et al. CAMEL: Communicative agents for "mind" exploration of large scale language model society, 2023a. URL https://arxiv.org/abs/2303.17760.

Ming Li et al. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models, 2023b. URL https://aclanthology.org/2023.emnlp-main.557/.

Yang Liu et al. G-Eval: NLG evaluation using GPT-4 with better human alignment, 2023. URL https://arxiv.org/abs/2303.16634.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Shunyu Yao, et al. Self-refine: Iterative refinement with self-feedback, 2023. URL https://arxiv.org/abs/2303.17651.

Reiichiro Nakano, Jacob Hilton, Aviral Kumar Arun, Mathis Prabhu, Michael Pei, et al. Webgpt: Browser-assisted question-answering with human feedback. In *arXiv*, 2021. URL https://openai.com/research/webgpt.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. NeurIPS 2022 (preprint), 2022. URL https://openai.com/research/instruction-following.

Bhargavi Paranjape et al. Artificial reasoning traces: Beyond the accuracy–explainability trade-off in NLP. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023. URL https://aclanthology.org/2023.acl-long.23/.

Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior, 2023. URL https://arxiv.org/abs/2304.03442.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Tania Lombrozo, Luke Zettlemoyer, Nenad Tomasev, and Douwe Kiela. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/df0a7b16c3fdbbc7aebca94c5e4a4c3f-Abstract.html.

Noah Shinn, Becky Labash, Ashwin Gopinath, Chris El-Sayed, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023. URL https://arxiv.org/abs/2303.11366.

Sainbayar Sukhbaatar, Rob Fergus, et al. Learning multiagent communication with backpropagation. In *Advances in Neural Information Processing Systems*, 2016. URL https://proceedings.neurips.cc/paper_files/paper/2016/hash/55b37c5c270e5d0f9e7c8c5d4e1e47a6-Abstract.html.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and verification. In *Proceedings of NAACL-HLT*, 2018. URL https://aclanthology.org/N18-1074/.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models, 2023. URL https://arxiv.org/abs/2302.13971.

Wei Wang et al. Large language models as judges: A survey, 2023. URL https://arxiv.org/abs/2310.19736.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models, 2022. URL https://arxiv.org/abs/2201.11903.

Qingyun Wu et al. Autogen: Enabling next-gen LLM applications via multi-agent conversation, 2023. URL https://arxiv.org/abs/2308.08155.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018. URL https://arxiv.org/abs/1809.09600.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models, 2023. URL https://arxiv.org/abs/2210.03629.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Soheil Zhuang, Zhuohan Lin, Zi Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena, 2023. URL https://arxiv.org/abs/2306.05685.

Jingyu Zhou et al. A survey on factuality in large language models, 2023. URL https://arxiv.org/abs/2310.05922.

A. Reproducibility Checklist

This appendix contains prompt templates, output contracts, and evaluation details sufficient to reproduce results without depending on a specific LLM vendor.

B. Output Contracts and Scoring

B.1. Task output schema

All methods return a structured JSON object with keys `final_answer`, `reasoning_steps`, and `supporting_facts`. Supporting facts are a list of {`title`, `sent_id`} pairs that reference provided context sentences, following the dataset's evidence format (Yang et al., 2018).

B.2. Strict vs canonicalized answer scoring

To ensure transparency, we report both strict answer scoring (minimal normalization) and canonicalized scoring (question-type canonicalization for yes/no and numeric outputs). This separation is important because answer-span formatting can inflate scores without improving grounding.

C. Prompt Templates

C.1. Peer critique template

```
You are a Constitutional Critic. Evaluate
    the message against:
Honesty, Fairness, Safety, Efficiency,
    Competence.
Return either:
COMPLIANT
or
VIOLATION: [Principle] - [Reason]
```

D. LLM-Judge Logical Consistency

The logical consistency judge evaluates whether reasoning steps support the final answer using only cited evidence sentences, returning a JSON verdict with fields `valid` and `score` (Zheng et al., 2023; Liu et al., 2023).